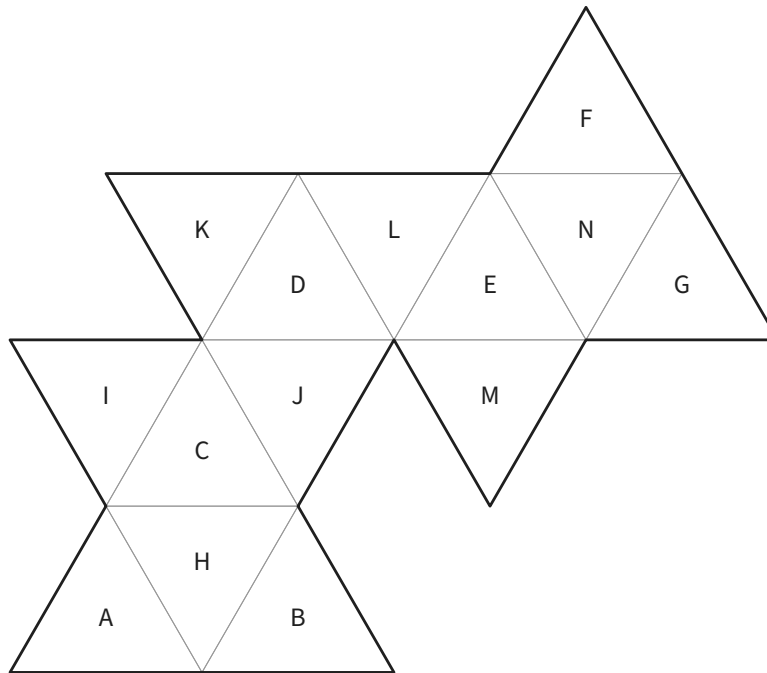


Enough partners?

Use small ($1\times$) blue rhombi on pairs of whole triangle spaces. No overlaps or pieces outside. Greens may fill the spaces you cannot pair.

Try it. Then explain it.

Count the up and down spaces. Can blues fill this board? If not, find the **fewest green fillers** needed to finish it.



My best filling: _____ blues and _____ greens.

Show a construction and a reason nobody can use fewer greens.

Look closer: Can several up spaces be competing for too few down partners? Mark a group and all its possible partners.

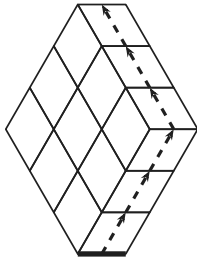
Go further: Turn your reason into a rule for any board with equal up/down counts. Would adding purple chevrons change your bound?

—————
One small block edge

Adult: print **100% / Actual Size**, single-sided. Default edge: 1 inch (25.4 mm). Compare with a $1\times$ green triangle before using these mats.

How far apart are two tilings?

Use the large board on page 5 with **15 small blues**, or use three R slips and three L slips. One move swaps neighboring unlike letters: $RL \leftrightarrow LR$. Keep the board fixed.



Start at the bold bottom edge. Cross each rhombus to its opposite horizontal edge. Each crossing goes right (R) or left (L). This ribbon reads RRRLLL, bottom to top.

1. Count the possibilities.

How many orders of three R's and three L's are there? Organize your list so nothing is missed or counted twice. Each order gives exactly one filling of this board.

My count and why the list is complete:

2. Get from one extreme to the other.

Find the fewest moves from RRRLLL to LLLRRR. Explain both why your route works and why no shorter route can work.

A best route, and a lower bound:

3. A trap for a promising shortcut.

Give a word one point for every L-before-R pair, including pairs that are not neighbors. Compare LRRRLL and RRLLLR: what are their scores? What is their actual distance?

Test a route. Can equal scores hide a positive distance?

Go further: Number the L's from left to right. Track their positions. Can you calculate the distance between *any* two words with the same numbers of R's and L's?

Build first. Explain by pointing, talking, or drawing.

How many best routes?

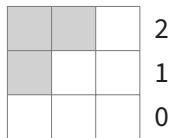
Use three R slips and three L slips. Go from RRRLLL to LLLRRR using the fewest moves. Routes are different if their sequence of words is different.

1. Find two routes.

Each L must cross each R. A shortest route makes each of these nine crossings once. Number the L's from left to right; their order never changes.

2. Keep a smaller record.

Record how many R's each L has crossed. For example, $(2, 1, 0)$ means the first L has crossed two R's, the second one, and the third none.



Shade left-justified rows in a 3×3 square. Row lengths must satisfy $3 \geq x_1 \geq x_2 \geq x_3 \geq 0$. Every move adds one square while keeping that rule. Start $(0, 0, 0)$; finish $(3, 3, 3)$.

3. Count without listing every route.

Add the route counts of all legal previous records. For $(2, 1, 0)$, use $(1, 1, 0)$ and $(2, 0, 0)$: $1 + 1 = 2$ routes. Work **down the left half**, then **down the right half**.

Record	Routes	Record	Routes
$(0, 0, 0)$	1	$(2, 2, 1)$	_____
$(1, 0, 0)$	1	$(3, 1, 1)$	_____
$(1, 1, 0)$	_____	$(3, 2, 0)$	_____
$(2, 0, 0)$	_____	$(2, 2, 2)$	_____
$(1, 1, 1)$	_____	$(3, 2, 1)$	_____
$(2, 1, 0)$	2	$(3, 3, 0)$	_____
$(3, 0, 0)$	_____	$(3, 2, 2)$	_____
$(2, 1, 1)$	_____	$(3, 3, 1)$	_____
$(2, 2, 0)$	_____	$(3, 3, 2)$	_____
$(3, 1, 0)$	_____	$(3, 3, 3)$	_____

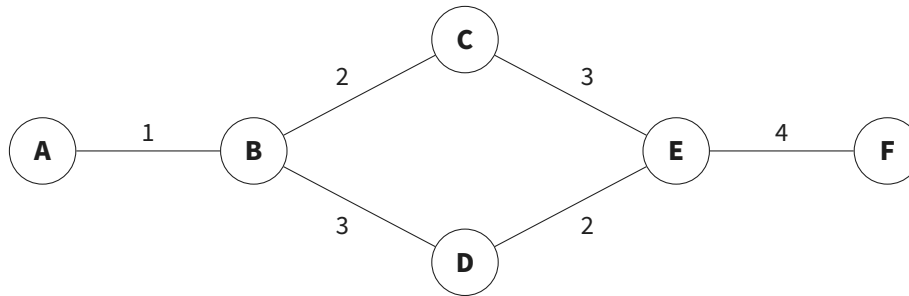
Why does adding the previous counts count each route exactly once?

Go further: Repeat with two R's and four L's. Is the number of shortest routes the same as the number of possible words?

Build first. Explain by pointing, talking, or drawing.

Does random mean equally often?

The six nodes below are the six tilings from the earlier activity. A line is one legal flip. Use a counter on the graph; you do not need to rebuild the blocks each turn.



1. Choose a legal move at random.

Start at A. At every turn choose each neighboring node with equal chance. Use the randomizer rule below. Predict which nodes you will visit most, then try **30 turns**. Tally your node *after* each turn; do not tally the initial A.

One neighbor: move there. Two: odd/even on a die chooses the alphabetically first/second neighbor. Three: reroll 4–6, then 1, 2, 3 chooses the first, second, third neighbor. Rerolls do not count as turns.

Node	A	B	C	D	E	F
Tally						

2. Explain a possible imbalance.

Write one unit at every node on scrap paper. In a *sharing model*, split each node's amount equally among its neighbors, all at once. Does every node receive one? Can you find starting amounts that come back unchanged after sharing? Fractions are allowed.

Stable starting amounts, and how I checked them:

3. Change the random rule.

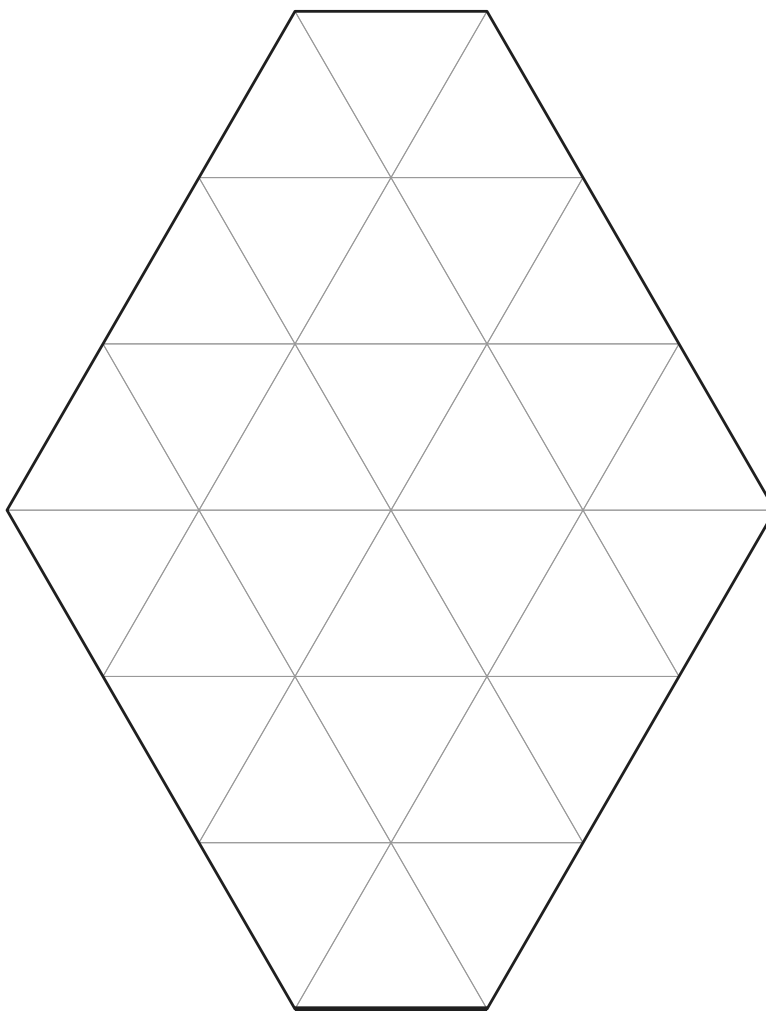
Make four identical folded slips labeled 1–4. Draw one, replace it, and mix again. Take the line with that label if it touches your node. Otherwise **stay put and count the turn**. Could this rule give equal long-run visit frequencies? Explain using both directions along a line.

Go further: Starting at A, count turns until your first return, then start a new trial. Compare the rules. Under rule 3 a stay at A on turn 1 counts as a return. What might the averages approach?

Build first. Explain by pointing, talking, or drawing.

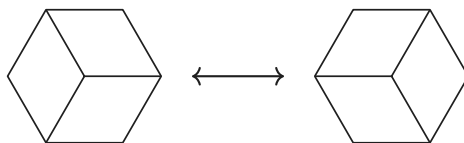
A larger world of tilings

Fill with **15 small (1×) blue rhombi**. Boundary side lengths: 1, 3, 3, 1, 3, 3 small edges. Keep the bold bottom edge fixed.



start the ribbon at this edge

One flip: Find three blues filling a small regular hexagon. Refill just that hexagon in its other way. On the ribbon this swaps one neighboring RL or LR.



—
One small block edge

Adult: print **100% / Actual Size**, single-sided. Default edge: 1 inch (25.4 mm). Compare with a **1×** green triangle before using these mats.